Junior Makers · Class 19

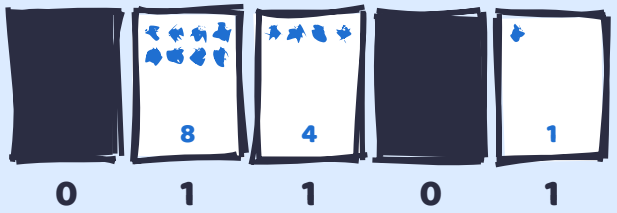
Binary Bracelets & Secret Codes

Speak the language of computers: 0 and 1

Bits | Binary numbers | Letter codes | Ciphers



A = 01000001



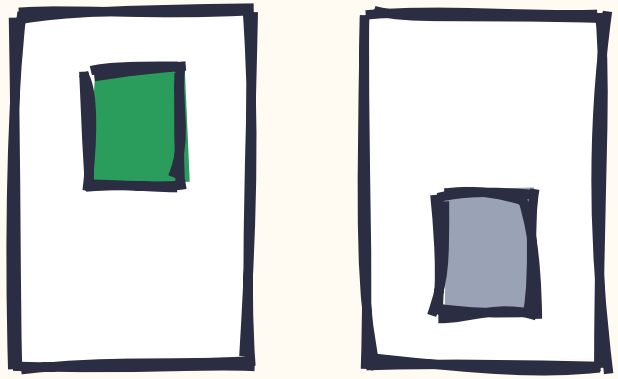
Did you know?
Every photo, song and message on your phone is stored as billions of 0s and 1s.

Let's think...

How does a phone store the letter A?

Only two things

Inside a computer chip there are billions of tiny switches. Each one can only be **ON** or **OFF**. That's all. Yet computers store your photos, songs and messages. How can just two states store **everything**?



1 = ON

0 = OFF

Tiny letters inside?

Magic?

Patterns of ON/OFF!

Today: you learn **binary**, wear your initial as a **binary bracelet**, and send **secret coded messages**.

Learning objectives

By the end of this class I can...

1	**Bits:** explain why computers store everything as **0s and 1s**	**Proof:** explain 'bit' with a light switch example
2	**Binary numbers:** convert numbers from **0 to 31** into binary with place-value cards	**Proof:** I can show 13 and 22 in binary
3	**Letters:** encode my initial as an **8-bit code** and make a bracelet	**Proof:** my bracelet shows my initial in 8 beads
4	**Codes & checks:** send a secret message with a **Caesar cipher** and catch errors with a **parity bit**	**Proof:** my partner decoded my message; I found the flipped card

Objective 1 · Bits · video

Video time!



The Basics of Binary for Kids

Potato Pirates · about 3 minutes · youtu.be/2Z03p3yhGmk

Watch for…

1. Why do computers use only **two** digits?
2. What are the place values in binary: 1, 2, 4, …?
3. How do you turn a binary number back into a normal number?

Scan to watch

After the video: Hold up fingers: down = 0, up = 1. Can your partner read your hand as a binary number?

Objective 1 · Bits

A bit is one switch: 0 or 1

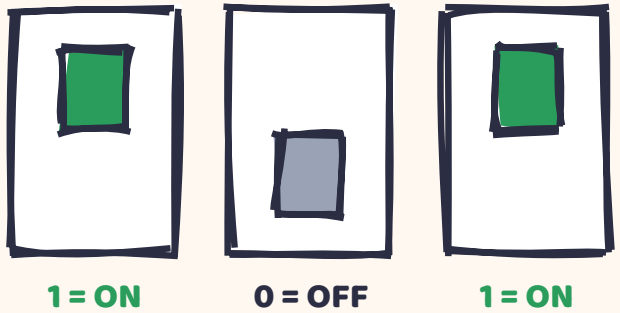
The idea

Bit = binary digit: **0** (off) or **1** (on).
One bit: 2 choices. Two bits: 4 choices (00, 01, 10, 11). Three bits: 8 choices. **Every extra bit doubles** the choices!

8 bits = 1 byte = 256 choices. Enough for every letter, number and symbol on your keyboard.

Why 2?
Computers COULD use 10 levels, but it's much easier to tell 'on' from 'off' reliably than to tell 10 levels apart.

Example: a light switch



A light switch is either on or off, never 'a bit on'. Electricity in a chip is the same: there or not there. That's why computers count in 2s, not 10s like us.

Objective 2 · Binary numbers

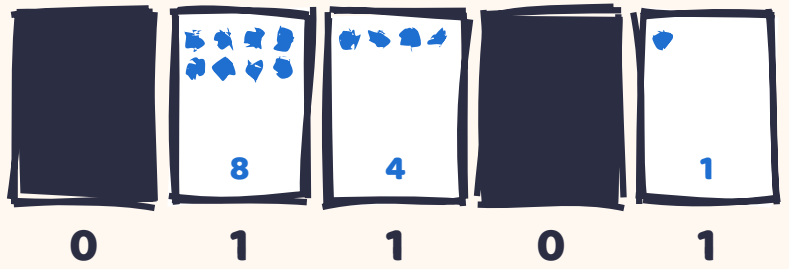
Binary place values: 16, 8, 4, 2, 1

The idea

In normal numbers, places are ones, tens, hundreds (×10). In binary they **double**: 1, 2, 4, 8, 16…

To read **01101**: add the places with a 1 → 8 + 4 + 1 = **13**.
To write **22**: take the biggest card that fits (16), left 6, then 4, left 2, then 2 → **10110**.

Example: paying with coins



To pay Rs 13 with coins of 8, 4, 2 and 1 (just one of each), you choose 8 + 4 + 1. Each coin is either used (1) or not (0). That's binary!

Objective 3 · Letters

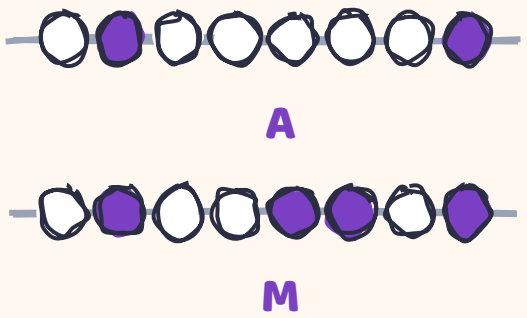
Letters are numbers too

The idea

Computers give every letter a number. Capital **A = 65**, B = 66, C = 67 … **Z = 90**. This list is called **ASCII**.

In 8 bits, every capital starts with **010**. The last 5 bits are the letter's place in the alphabet:
A (1st) → 010 **00001** · **M** (13th) → 010 **01101** · **Z** (26th) → 010 **11010**

Example: a secret classroom code



If your class agrees 'Ali = 1, Sara = 2, Bilal = 3', you can pass a note saying just '3' and everyone knows it means Bilal. ASCII is a worldwide agreement like that, for letters.

LEARN

Real world

Binary is hiding everywhere



Messages
Every letter you type becomes 8 or more bits



Photos
Each pixel's colour is stored as 24 bits



Music
Sound waves are measured 44,100 times a second, each as bits



QR codes
Black and white squares are 1s and 0s

? A photo has 12 million pixels × 24 bits. How many bits is that? (Use a calculator!)

Answer
288 million bits = 36 million bytes ≈ 36 MB, before compression.

Objective 4 · Codes & checks

Secret codes: the Caesar cipher

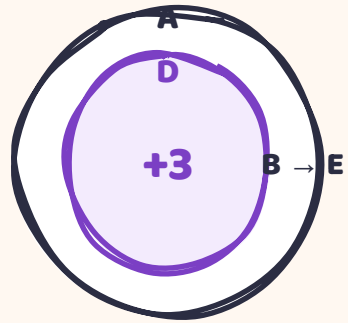
The idea

Julius Caesar hid messages by **shifting** every letter along the alphabet. With a shift of **3**: A → D, B → E, C → F … X → A, Y → B, Z → C.

HELLO becomes **KHOOR**. Only someone who knows the **key** (3) can shift back and read it.

Weak!
A shift cipher is easy to break: there are only 25 keys to try. Real security uses much harder maths.

Example: a secret handshake



Your team's secret handshake only works if everyone knows the moves. The key to a cipher is like that: share it with friends, never with the 'enemy'.

Objective 4 · Codes & checks

Catch mistakes with a parity bit

The idea

Messages sometimes get damaged: a 1 flips to a 0. To spot it, add one extra bit so the number of 1s is always **even**. 01000001 has two 1s → add **0**. 01000011 has three 1s → add **1**.

If a message arrives with an **odd** number of 1s, the computer knows something went wrong and asks again.

Example: counting the class on a trip



C has three 1s → add 1 → four (even)

The teacher counts 30 students before and after the bus ride. If the count is 29, she knows something is wrong, even before knowing who is missing. Parity is a quick count that catches errors.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human binary
2. Binary speed challenge
3. Make your binary bracelet
4. Read the bracelets
5. Send a secret message
6. Parity magic trick
7. Code-breaker challenges

MAKE

Class game

Human binary

How to play

1. Five students stand in a line holding cards: **16, 8, 4, 2, 1** (dots on the front).
2. Card facing the class = 1. Card turned away = 0.
3. Teacher calls a number: **5!** The team flips cards to show it (00101).
4. Then: 10, 7, 19, 31… What's the biggest number possible?
5. Last round: the class shows a binary number, the team says it.

Then discuss

What pattern do you see in 1, 2, 4, 8, 16? What would the next card be?



Big idea

Each card is double the one to its right. That's binary place value.

Objective 2 · Binary numbers · practise

Binary speed challenge

Number	Binary (16 8 4 2 1)
5	
9	
13	
22	
31	

Binary	Number
00011	
01010	
10000	
10101	
11111	

Answers

Cover! 00101, 01001, 01101, 10110, 11111 · 3, 10, 16, 21, 31

Race your partner: first to finish both columns correctly wins. Use the cards if you need them!

MAKE

Objective 3 · Build step 1 of 2

Make your binary bracelet

a Find your initial's place in the alphabet (e.g. S = 19th).

b Write it in 5 bits with your cards: 19 = 16 + 2 + 1 → **10011**.

c Put **010** in front: S = **010 10011**.

d Colour the template: **1 = dark bead**, **0 = light bead**. 8 beads in a row.

e Cut it out and tape it round your wrist (or thread real beads at home!).



S = 01010011

Check: Your partner can read your bracelet and say your initial.

Objective 3 · Letters · decode

Read the bracelets

Bracelet	Last 5 bits → number	Letter
01000011		
01001111		
01000100		
01000101		

Hidden word

Read the 4 bracelets in order. They spell a word you know!

Answer

Cover! 3 C · 15 O · 4 D · 5 E → CODE. Now write your own 4-letter word for a partner.

MAKE

Objective 4 · Build step 2 of 2

Send a secret message

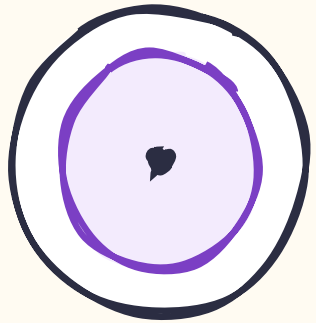
a Make a **cipher wheel** from the template: two paper circles and a split pin.

b Agree a secret key with your partner (a number from 1 to 25).

c Write a short message (5 words max) and encode it.

d Swap. Decode your partner's message by shifting **back**.

e Bonus: crack another team's message WITHOUT the key. How long did it take?



Check: Your partner reads your message correctly.

Class game

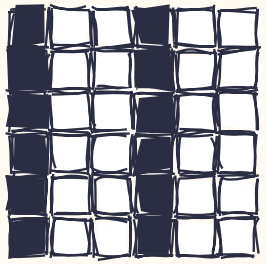
Parity magic trick

How to play

1. Lay out a 5 × 5 grid of cards, randomly black or white side up.
2. The 'magician' adds a 6th row and column 'to make it harder', secretly making every row and column have an **even** number of black cards.
3. The magician turns away. A volunteer flips ONE card.
4. The magician turns back and instantly finds the flipped card!
5. How? Find the row and column with an odd count.

Then discuss

Explain the trick to your partner. Then try being the magician yourself!



Big idea

Computers use this exact idea to fix errors in QR codes, Wi-Fi and hard drives.

Level up!

Code-breaker challenges

Choose your challenge



Big numbers: add a 32 card. What's the biggest number with 6 bits?



Full name: make a bracelet for your whole first name.



Crack it: decode 'DQG WKHQ' with no key. Which shift works?



Count on fingers: binary on 10 fingers goes up to 1023!

Think like an engineer

Why do computers use bytes of 8 bits and not 5?

Boss level

Invent your own secret code that is harder to crack than Caesar's. Can a partner crack it?

Real-life secret codes

Codes that keep you safe

Encryption

When you see a **padlock** 🔒 in your browser, your messages are scrambled with codes far stronger than Caesar's. Only the website can unscramble them.

Passwords

A secret code is only safe if the KEY is secret. The same is true for passwords: never share them, even with your best friend.

Objective check

Can I really do it? Prove it.

01	What is a bit? Why do computers use only 0 and 1?	○ ○ ○
02	Show 13 and 22 in binary using the cards.	○ ○ ○
03	Show your bracelet. Ask a partner to read your initial.	○ ○ ○
04	Read out your secret message, then let your partner decode it. Explain how parity catches a mistake.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Bracelet parade: everyone reads someone else's bracelet out loud.

Show what you know

What did we learn?

- [x] A bit is 0 or 1; 8 bits make a byte
- [x] Binary places double: 1, 2, 4, 8, 16…
- [x] Letters are numbers: A = 65 = 01000001
- [x] A Caesar cipher shifts letters by a secret key
- [x] A parity bit helps catch errors

At home

Teach someone at home to count to 31 on one hand in binary!



Exit ticket: what number is 10101 in binary?

Next class: Joystick Maze Game!